

Chapter 10 — Buildings, Paintings and Books

Art, Architecture, Literature and Science

The one-line point: This is **YOUR MAINS CHAPTER**. The GS1 syllabus line — "*salient aspects of Art Forms, Literature and Architecture from ancient to modern times*" — is answered from exactly this material. It is also the biggest single bucket of Prelims questions in the whole book.

← Ch.9 | Index |  **Final chapter**

Syllabus Classification

Field	Value
Subject	History → Ancient India → ★ ART & CULTURE
Sub-discipline	Art history, architecture, literature, science & technology
Period	~3rd century BCE – 8th century CE (cuts across Ch.6–9)
Prelims (GS Paper I)	★★★★ <i>Art & Culture</i> — the single largest question bucket in ancient India (8–12 Qs a year when combined with medieval)
Mains (GS Paper I)	★★★★★ THE core chapter . The syllabus line on art, literature and architecture is answered from here
Also feeds	Science & Tech (Aryabhata, zero, metallurgy), Polity (national symbols), IR (UNESCO sites, cultural diplomacy), Environment (ancient water engineering), Essay (India's scientific and cultural heritage)
Weightage	★★★★★ — the highest-value chapter in the book, jointly with Ch.3 and Ch.6
Overlaps with	Everything. It is the cultural output of Ch.6–9

 **This is where "Ancient History" becomes "Art & Culture."** If your time is short, this chapter and Chapter 6 give you the best return per hour in the entire book.

Learning Outcomes

By the end of this chapter you should be able to:

- ☐ **Label a stupa diagram** — *anda, harmika, yashti, chhatra, medhi, pradakshina patha, vedika, torana*.
- ☐ **Describe** Sanchi, Bharhut and Amaravati, and say who built and enlarged each.
- ☐ **Classify** cave architecture — *chaitya* vs *vihara* — and name the key sites.
- ☐ **Describe** Ajanta and Ellora precisely, including what makes Ellora exceptional.
- ☐ **Label a temple diagram** — *garbhagriha, shikhara, mandapa, vimana, gopuram, amalaka, kalasha*.
- ☐ **Compare** the **Nagara**, **Dravida** and **Vesara** styles.
- ☐ **Explain** the Mahabalipuram rathas and the Kailashnatha temple at Ellora.
- ☐ **Name** the major Sanskrit and Tamil works with authors.
- ☐ **Distinguish** the two epics, the Puranas, and the Jataka/Panchatantra tradition.

- ☐ **State** Aryabhata's specific contributions and explain the significance of **zero** and the **decimal place-value system**.
- ☐ **Name** the key figures in medicine, astronomy and metallurgy.
- ☐ **Explain** the **Mehrauli Iron Pillar** and why it does not rust.
- ☐ **List** the ancient-India **UNESCO World Heritage Sites**.

Start Here: Why This Chapter Is Different

Every previous chapter asked "*what happened?*" This one asks "*what did they MAKE?*" — and the things they made are still standing.

WHAT SURVIVES FROM ANCIENT INDIA THAT YOU CAN VISIT TODAY

SANCHI STUPA	- begun by Ashoka, ~2,300 years old
AJANTA CAVES	- paintings still vivid after 1,500 yrs
ELLORA KAILASHNATHA	- an entire TEMPLE carved DOWNWARD out of one rock. 200,000 tonnes removed
MAHABALIPURAM	- monolithic temples on a beach
MEHRAULI IRON PILLAR	- 1,600 years old, still barely rusted
SARNATH LION CAPITAL	- on every rupee note in your wallet
0 THE DIGIT ZERO	- in every calculation you have ever run, on every machine you have used

 **That last one deserves a moment.** Every binary representation, every integer type, every memory address in every program you have ever written **depends on positional notation with zero as a placeholder**. Without it, arithmetic is Roman-numeral arithmetic — and there is no practical computation, no algebra, no computer science.

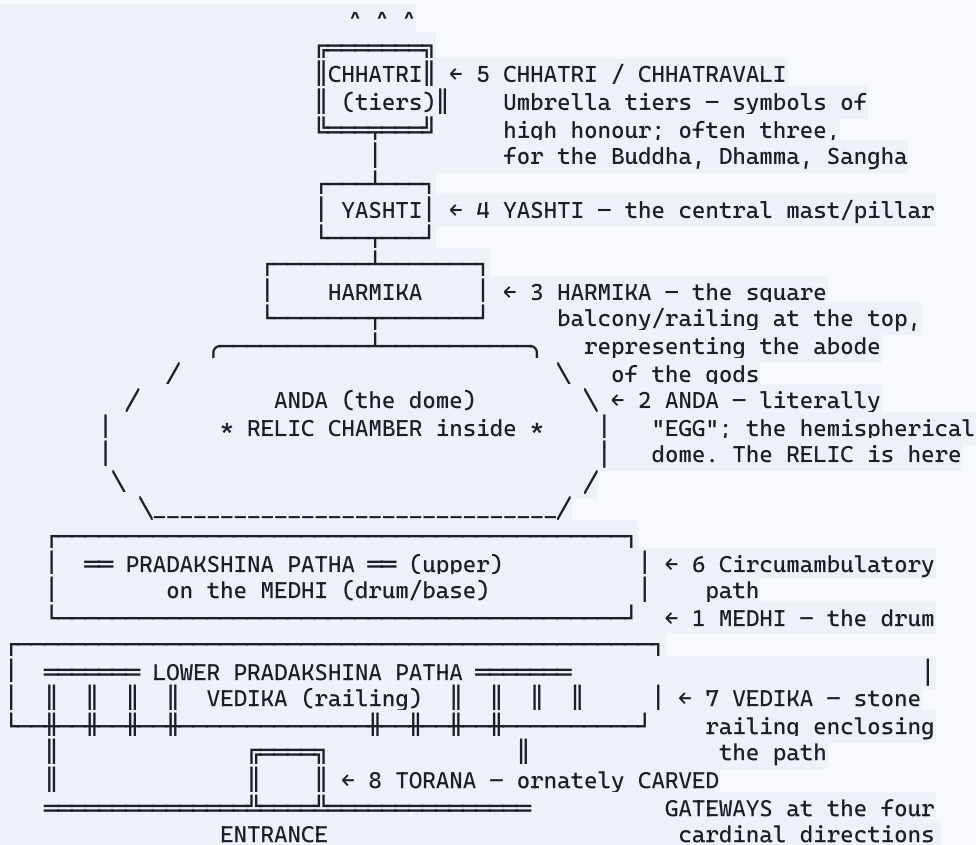
The most consequential thing in this chapter is not a building. It's a digit.

PART A — BUILDINGS

1 STUPAS — Buddhist Architecture

A **STUPA** (Sanskrit: "mound") is a **mound-shaped structure enshrining a relic** — of the Buddha, a revered monk, or a sacred object. It is not a building you enter; it is an object you **walk around**.

★ ★ ★ Anatomy of a Stupa — draw and label this



Part	What it is
Anda ★	The hemispherical dome ("egg") containing the relic chamber
Medhi	The drum/base on which the anda rests
Harmika ★	The square railing/balcony atop the dome — the abode of the gods
Yashti ★	The central mast rising from the harmika
Chhatri / Chhatravali ★	The umbrella tiers on the yashti — symbols of honour
Pradakshina patha ★	The circumambulatory path — devotees walk clockwise around the stupa
Vedika ★	The stone railing enclosing the path
Torana ★ ★	The elaborately carved gateways at the four directions

★ **Why walk around it?** Circumambulation (*pradakshina*) is the act of worship — you keep the sacred object on your **right**. The stupa is designed to be **experienced in motion**, and the carved railings and toranas unfold as a **visual narrative** as you walk. **That's a genuinely good point for a Mains answer on stupa art.**

The great stupas

Stupa	Location	Built/Enlarged by	Famous for
SANCHI ★★ ★	Madhya Pradesh	Ashoka (original brick core) → enlarged in stone by the SHUNGAS → toranas added by the SATAVAHANAS	★ The best-preserved stupa in India ; magnificent carved toranas with Jataka scenes; UNESCO World Heritage Site (1989) . Also has an Ashokan pillar. Saved from demolition and restored under John Marshall
BHARHUT ★ ★	Madhya Pradesh	Shunga period	★ Earliest extensive narrative relief sculpture ; railings and gateway now largely in the Indian Museum, Kolkata . Figures are flat, low-relief, with inscribed labels naming the scenes
AMARAVATI ★ ★	Andhra Pradesh (on the Krishna)	Satavahanas	★ White marble/limestone ; slender, dynamic, crowded, highly expressive figures; narrative panels ; influenced Sri Lankan and Southeast Asian art . Much of it is now in the Government Museum Chennai and the British Museum
Sarnath (Dhamek)	Uttar Pradesh	Gupta period (on Ashokan foundations)	Marks the site of the first sermon
Nagarjunakonda	Andhra Pradesh	Ikshvakus	Important Buddhist site, largely relocated due to a dam
Bodh Gaya (Mahabodhi)	Bihar	Ashoka onward	UNESCO site ; the Bodhi tree

★★ **THE ANICONIC POINT** — this is the single most-asked idea in early Buddhist art (and the basis of **UPSC Mains 2016**).

For roughly the first four centuries, **Buddhist art did NOT depict the Buddha in human form**. He was indicated by **symbols**:

Symbol	Represents
Empty throne	The Buddha's presence / enlightenment
Footprints (<i>buddhapada</i>)	His presence and his path
Bodhi tree	The Enlightenment at Bodh Gaya
Wheel (<i>dharmachakra</i>)	The First Sermon at Sarnath
Stupa	The Mahaparinirvana at Kushinagar
Riderless horse	The Great Departure from Kapilavastu
Lotus	Birth / purity

Human images of the Buddha first appear ~1st century CE in the Gandhara and Mathura schools (Ch.8).

🌀 **Note the elegance of the design**: each symbol maps to one great event, so a viewer walking around the railing reads the Buddha's whole life without a single depiction of his body. **Absence used as a device to convey transcendence**. That sentence, in a Mains answer, is worth real marks.

2 🏡 CAVE ARCHITECTURE — Rock-Cut Wonders

Indian rock-cut architecture is **subtractive** — you don't build up, you **carve away**. There is no scaffolding, no assembly, and **no possibility of correcting a mistake**.



This is the point that makes rock-cut architecture astonishing. It is a **write-once, no-undo medium**. The entire structure — pillars, brackets, sculptures, ceilings, the whole spatial plan — had to be **fully resolved in advance** and executed **top-down**, because you cannot add stone back. Compare that to masonry construction, where you can dismantle and rebuild a course. **There is no rollback here.** And they did it at the scale of Kailashnatha.

Two types — know the distinction ★★

	CHAITYA ★	VIHARA ★
What it is	A prayer/worship hall	A monastery / residence for monks
Plan	Apsidal (rounded at the far end), with a nave and two side aisles	Square/rectangular hall with small cells cut around its sides
Contains	A STUPA at the apsidal end	Cells for sleeping; later, a shrine with a Buddha image
Ceiling	Barrel-vaulted , often with stone ribs imitating wooden beams ★	Flat
Facade	A large horseshoe-shaped chaitya window (<i>gavaksha</i>) ★ letting light onto the stupa	Simpler
Best examples	★ KARLE (the largest and finest chaitya), Bhaja, Bedsa, Ajanta caves 9, 10, 19, 26	Ajanta, Nasik, Ellora, Bagh

★ **The wooden imitation detail matters.** Early chaityas have **stone ribs carved to look like wooden beams**, and some even had **actual wooden ribs fitted**. This proves the form was **translated from an earlier wooden architecture that has not survived** — exactly Chapter 1's point about organic material rotting away. **The same evidence appears in the Lomas Rishi cave facade (Ch.7).**

🎨 AJANTA — the paintings ★★★

Aspect	Detail
Where	Maharashtra , in a horseshoe-shaped gorge on the Waghora river
Number	30 caves (some unfinished)
Period	Two phases — Satavahana/Hinayana (c. 2nd century BCE – 1st century CE) and Vakataka/Mahayana (c. 5th–6th century CE, the great painting phase)
Patrons	Satavahanas , then the Vakatakas (notably under Harishena — a <i>different</i> Harishena from Ch.9's poet)
Type	Both chaityas and viharas
★ FAMOUS FOR	PAINTINGS — the finest surviving ancient Indian murals
Subjects	JATAKA tales , the Buddha's life, Bodhisattvas, and vivid scenes of court and everyday life
★ Iconic works	Padmapani (Bodhisattva with a lotus) and Vajrapani (with a thunderbolt), in Cave 1 ; the Dying Princess ; Mahajanaka Jataka
★ TECHNIQUE	TEMPERA on a prepared plaster ground — ⚠️ NOT true fresco . In true fresco, pigment is applied to wet plaster; at Ajanta the plaster was allowed to dry and pigment applied with a binding medium. This distinction is asked
Pigments	Natural minerals — red and yellow ochre, lampblack, lime white, lapis lazuli (blue — imported!), terre verte
Rediscovered	1819 , by a British hunting party (John Smith)
Status	UNESCO World Heritage Site (1983)

★ **Why the paintings survived:** the caves were **abandoned and forgotten**, and the deep gorge kept them dark and dry. **Neglect preserved them.** (Their main threats today are humidity and visitor traffic.)

🏞️ ELLORA — the greatest of all ★★ ★

Aspect	Detail
Where	Maharashtra , near Aurangabad (~100 km from Ajanta)
Number	34 caves
Period	c. 6th–10th century CE
★★ THE KEY FACT	★ THREE RELIGIONS SIDE BY SIDE: Buddhist (caves 1–12), Hindu (13–29), Jain (30–34) — carved over centuries in the same cliff
★★★ CAVE 16	The KAILASHNATHA TEMPLE — see below
Patrons	Chiefly the RASHTRAKUTAS (Kailashnatha under KRISHNA I , 8th century); also Chalukyas and Yadavas
Other notable caves	Cave 10 (Vishvakarma — a Buddhist chaitya), Cave 15 (Dashavatara), Cave 21 (Ramesvara), Cave 32 (Indra Sabha — Jain)
Status	UNESCO World Heritage Site (1983)

🌀 **Ellora's religious coexistence is a superb Mains and Essay point.** Buddhist, Hindu and Jain monuments carved into a **single cliff face over four centuries**, under successive dynasties, without any being destroyed. **It is physical evidence of religious pluralism in practice** — far more persuasive than an abstract assertion. **Use it in answers on India's tradition of tolerance.**

🏞️ The Kailashnatha Temple (Ellora Cave 16) ★★ ★

Arguably the single most astonishing structure in India.

HOW IT WAS BUILT – the part that should stop you

NORMAL BUILDING:

Start at the bottom,
add stone upward.

▲
| add
| add
| add

GROUND

You can dismantle and
rebuild a course.

KAILASHNATHA:

Start at the TOP of the hill.
Cut three deep trenches **DOWNWARD**
to isolate a block of rock.
Then carve the entire temple –
towers, pillars, halls, courtyards,
elephants, sculptures – **OUT** of that
single block, working **DOWNWARD**.

▼ carve
▼ carve
▼ carve

BEDROCK

NO ADDING. NO REBUILDING.
NO MISTAKES PERMITTED.

Fact	Detail
Patron	KRISHNA I, Rashtrakuta dynasty, 8th century CE
Dedicated to	Shiva (modelled on Mount Kailash, his abode)
Method	★ Carved TOP-DOWN from a single rock — MONOLITHIC
Rock removed	Estimated ~ 200,000 tonnes
Size	~ 60 m long × 30 m wide × 30 m high
Features	Multi-storeyed <i>vimana</i> , <i>mandapa</i> , Nandi shrine, life-size elephants appearing to carry the temple on their backs , panels of the Ramayana and Mahabharata, and a famous Ravana shaking Mount Kailash relief
Style	Dravidian in idiom, though located in the Deccan

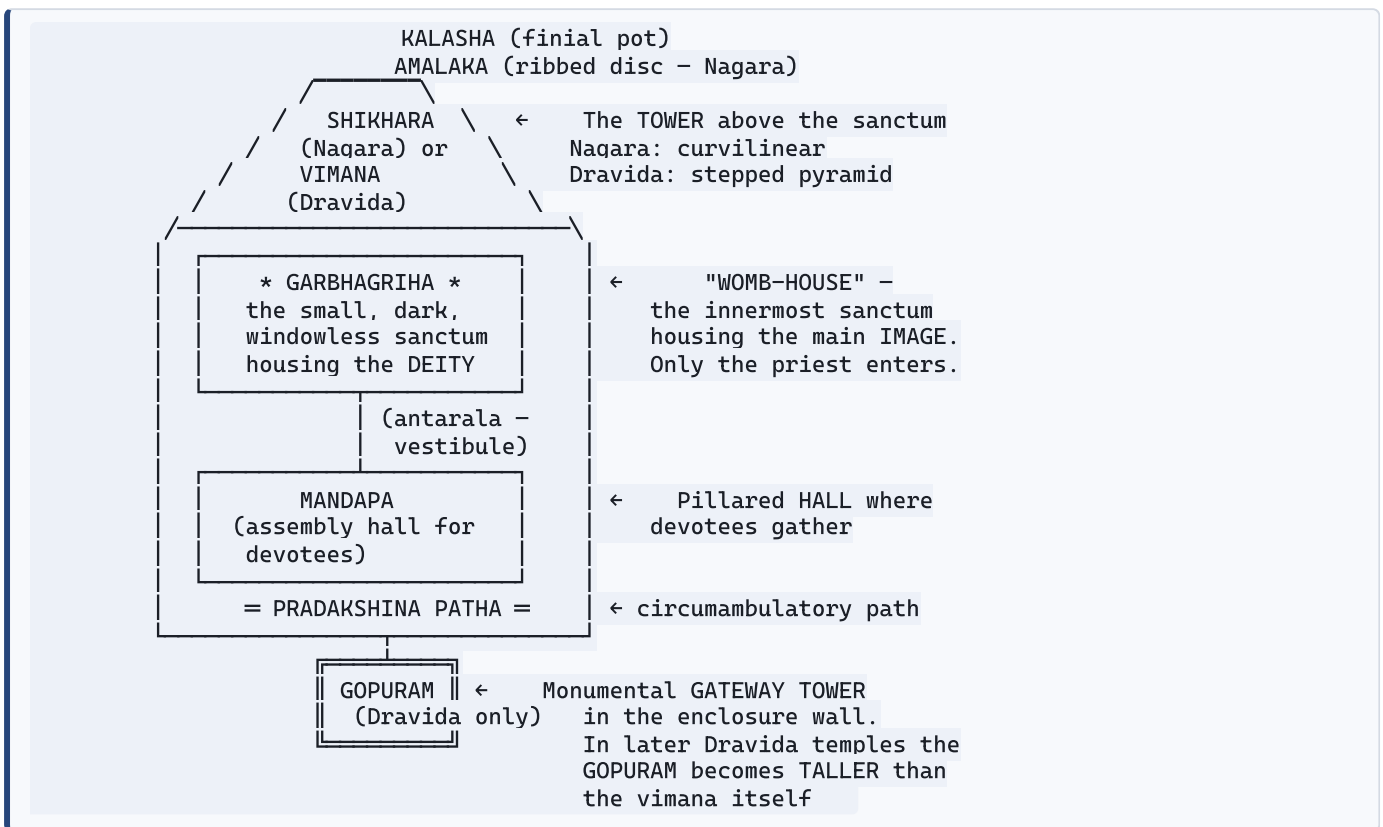
🎯 **The right way to say it in an answer:** *"Kailashnatha is not built architecture but sculpted architecture — a temple realised by subtraction rather than construction, requiring the entire design to be resolved before the first cut was made."* **That framing shows you understand the achievement, not just the statistic.**

Other cave sites to know: **Barabar** (Ch.7 — earliest, Ajivikas), **Karle** (largest chaitya), **Bhaja**, **Bedsa**, **Nasik**, **Kanheri** (Maharashtra), **Udayagiri–Khandagiri** (Odisha — **Jain**, under **Kharavela**, with the **Hathigumpha inscription**), **Bagh** (MP — paintings), **Elephanta** ★ (Maharashtra — **UNESCO**, famous **Trimurti/Sadashiva** sculpture), **Badami** (Chalukya).

3 🏛️ **TEMPLES — A New Architecture Is Born**

The rise of **Bhakti** (Ch.8) made **image worship in a permanent house of the deity** central — and the temple was invented to serve it.

★★ Anatomy of a temple



Part	Meaning
Garbhagriha ★★	"Womb-house" — the small dark sanctum housing the main image
Shikhara ★	The tower over the sanctum (Nagara term)
Vimana ★	The pyramidal tower over the sanctum (Dravida term)
Mandapa ★	The pillared hall for devotees
Antarala	Vestibule connecting mandapa and garbhagriha
Gopuram ★★	Monumental gateway tower — Dravida
Amalaka ★	Ribbed, disc-shaped stone crowning a Nagara shikhara
Kalasha	The pot-shaped finial at the very top
Shikhara vs Vimana ⚠	In Dravida , "shikhara" refers only to the crowning element on top of the vimana. This is a classic Prelims trap

★ ★ ★ Nagara vs Dravida vs Vesara

	NAGARA (North)	DRAVIDA (South)	VESARA (Deccan — hybrid)
Region	North India	Tamil Nadu, South	Deccan/Karnataka
Tower	SHIKHARA — curvilinear , beehive-shaped, tapering smoothly	VIMANA — stepped PYRAMID of receding storeys (<i>talas</i>)	Blend of both
Crowning element	AMALAKA + kalasha	Shikhara (a small octagonal/domed cupola) + kalasha	Mixed
Boundary wall	★ Usually ABSENT	★ PRESENT , with elaborate gopurams	Present
Gopuram	✗ No	★ YES — and increasingly dominant	Sometimes
Water tank	Usually absent	★ Present (temple tank)	Sometimes
Sub-styles	Latina, Phamsana, Valabhi; regional: Odisha, Khajuraho, Solanki	—	—
Examples	Khajuraho (Chandela), Konark & Lingaraja (Odisha), Dashavatara temple Deogarh	Brihadeshwara, Thanjavur (Chola); Shore Temple & rathas, Mahabalipuram; Kailasanatha, Kanchi (Pallava)	Pattadakal , Aihole, Badami (Chalukya); later Hoysala (Belur, Halebidu)

🧠 **The fastest discriminator: look for the GOPURAM and the boundary wall.** Present → **Dravida**. Absent, with a smooth curving tower topped by a ribbed **amalaka** → **Nagara**.

🌊 Mahabalipuram (Mamallapuram) — Pallava ★ ★ ★

UNESCO World Heritage Site. Named after **Narasimhavarman I "Mamalla"** (Ch.9).

Monument	Detail
PANCHA RATHAS ★ ★	Five MONOLITHIC temples , each carved from a single boulder, named after the Pandavas and Draupadi: Draupadi, Arjuna, Bhima, Dharmaraja (Yudhishtira), Nakula-Sahadeva . ★ Each is in a DIFFERENT architectural style — effectively an architectural sample book in stone , showing the experiments that produced the Dravida temple
SHORE TEMPLE ★ ★	A STRUCTURAL (built, not carved) temple by the sea, under Narasimhavarman II (Rajasimha) . ★ Marks the crucial transition from rock-cut to structural architecture
DESCENT OF THE GANGA / Arjuna's Penance ★ ★	An enormous open-air bas-relief (~30 m × 15 m) on a rock face — the largest such relief in the world — with gods, sages, animals and a famous family of elephants
Cave temples (mandapas)	Varaha, Mahishasuramardini, Krishna mandapas

★ **The Pallava sequence is the classic Mains illustration of architectural evolution:**
Mahendravarman I (rock-cut cave temples/*mandapas*) → **Narasimhavarman I "Mamalla"** (monolithic **rathas**) → **Narasimhavarman II "Rajasimha"** (**structural** temples — Shore Temple, Kailasanatha at Kanchi) → **Nandivarman** (later structural). **Cave** → **monolith** → **structural**. Learn this four-stage progression; it answers "trace the evolution of temple architecture" beautifully.

Other key temple sites

Site	Note
AIHOLE ★	Chalukya; called ★ "the cradle of Indian temple architecture" — an experimental ground with ~125 temples. Durga temple (apsidal, chaitya-derived), Lad Khan temple
PATTADAKAL ★★	Chalukya; ★ UNESCO World Heritage Site ; both Nagara and Dravida temples side by side — a genuine architectural laboratory. Virupaksha temple is the finest
BADAMI	Chalukya; rock-cut cave temples (Hindu and Jain)
Kailasanatha, Kanchipuram	Pallava structural temple (Rajasimha)
Brihadeshwara, Thanjavur ★★	CHOLA , built by RAJARAJA I (c. 1010 CE). ★ UNESCO ("Great Living Chola Temples"). One of the tallest vimanas in India
Deogarh (Dashavatara temple)	Gupta-period Nagara temple, UP
Sanchi Temple 17, Tigawa, Bhitargaon ★	Early Gupta temples — the beginnings of the free-standing temple ; Bhitargaon is brick with a shikhara

★ **Where temples begin:** the **Gupta period** produced India's **earliest surviving free-standing structural temples** — small, flat-roofed, simple (Sanchi Temple 17, Tigawa) — which then evolved shikharas (Deogarh, Bhitargaon). **The Guptas invented the temple form; the Pallavas, Chalukyas and Cholas perfected it.**

PART B — BOOKS

The Great Literature

The Two Epics ★★

	MAHABHARATA	RAMAYANA
Traditional author	VYASA	VALMIKI ★ (called the <i>Adi Kavya</i> — "the first poem")
Language	Sanskrit	Sanskrit
Length	★ ~100,000 verses — the longest poem in the world	~24,000 verses
Story	The Kaurava–Pandava war over the Kuru throne	Rama's exile, Sita's abduction by Ravana, the war in Lanka
Contains	★ The BHAGAVAD GITA ; the <i>Shanti Parva</i> ; countless embedded tales	—
Composition	Grew over ~1,000 years (c. 500 BCE – 400 CE)	c. 5th–4th century BCE onward
Tamil version	—	★ Kamban's Ramavataram (12th c.)

The Puranas ★

Purana = "old." Compiled largely in the **Gupta period** and after.

- **18 Mahapuranas** (Vishnu, Shiva, Bhagavata, Markandeya, Matsya, etc.)
- Contain: **myths, cosmology, genealogies of gods and kings (vamshanucharita)**, pilgrimage guides, ritual

- ★ **In Sanskrit, but meant to be heard by ALL — including women and Shudras**, who were barred from Vedic study. **Recited publicly by priests.** This is an important democratising point
- Historically useful for **dynastic lists**, though the chronology needs careful handling

Stories, Fables and Drama

Work	Detail
JATAKAS ★★	~550 stories of the Buddha's previous births. In Pali ; part of the <i>Sutta Pitaka</i> . ★ Carved on the stupa railings at Sanchi, Bharhut and Amaravati and painted at Ajanta — literature and art fused
PANCHATANTRA ★★	By VISHNU SHARMA (c. 3rd century CE). Animal fables teaching <i>nitī</i> (practical wisdom/statecraft). ★ One of the most widely translated books in world literature — travelled to Persia (<i>Kalila wa Dimna</i>), Arabic, Greek, Latin and every major European language. It is the ancestor of Aesop-style fable collections in the West
HITOPADESHA	By Narayana ; similar animal fables
KATHASARITSAGARA	By Somadeva (11th c.); "ocean of the streams of story"
Brihatkatha	By Gunadhya , in Paishachi Prakrit (lost, known through later versions)

KALIDASA ★★ ★ — India's greatest Sanskrit poet

Traditionally placed in the **Gupta court** (associated with **Chandragupta II Vikramaditya**, though the dating is debated).

Work	Type	Note
ABHIJNANA SHAKUNTALAM ★★	Play	The story of Shakuntala and Dushyanta . ★ Translated into English (1789 by William Jones) and German — Goethe was famously rapturous about it. India's most celebrated drama
MEGHADUTA ★★	Lyric poem	"The Cloud Messenger" — an exiled <i>yaksha</i> asks a passing cloud to carry a message to his wife in the Himalayas. ★ The poem traces the cloud's route, making it a poetic geography of India
RAGHUVAMSHA	Epic poem	The dynasty of Raghu (Rama's line)
KUMARASAMBHAVA	Epic poem	The birth of Kartikeya
RITUSAMHARA	Lyric poem	The six seasons
VIKRAMORVASHIYAM, MALAVIKAGNIMITRAM	Plays	The latter is set in the Shunga court

★ **Kalidasa's insight worth quoting in a Mains answer:** in his plays, **kings and brahmanas speak Sanskrit; women and lower-status characters speak Prakrit (Ch.9). The drama encodes the social hierarchy in its very language.**

Other Sanskrit authors

Author	Work
Shudraka ★	Mrichchhakatika ("The Little Clay Cart") — unusual for its ordinary, non-royal characters (a courtesan, a poor brahmana)
Vishakhadatta ★	Mudrarakshasa (Chandragupta & Chanakya), <i>Devichandraguptam</i>
Banabhatta ★	Harshacharita, Kadambari (one of the world's earliest novels)
Bhasa	Early dramatist — <i>Swapnavasavadatta, Urubhanga</i>
Bharavi	<i>Kiratarjuniya</i>
Dandin	<i>Dashakumaracharita, Kavyadarsha</i>
Ashvaghosha ★	Buddhacharita , <i>Saundarananda</i> (Kanishka's court, Ch.8)
BHARATA ★ ★	NATYASHASTRA — the foundational treatise on drama, dance, music and aesthetics ; source of the <i>rasa</i> theory. ★ The basis of all Indian classical dance forms — a very high-value Art & Culture item

🌿 Tamil literature (Ch.8 recap — but know the pairs cold)

Work	Author
Tolkappiyam	Tolkappiyar — earliest Tamil grammar
Tirukkural ★	Thiruvalluvar — 1,330 ethical couplets
Silappadikaram ★	Ilango Adigal
Manimekalai ★	Sattanar
Ettuthogai & Pattuppattu	Sangam anthologies

PART C — SCIENCE

5 ★ ★ ★ Mathematics and Astronomy

ARYABHATA (476–550 CE) ★ ★ ★

Wrote the **ARYABHATIYAM** at the age of 23, in Sanskrit verse.

Contribution	Detail
Earth rotates ★★	★ Stated that the EARTH ROTATES ON ITS AXIS , and that the apparent westward motion of the stars is caused by this rotation — a millennium before Copernicus
Cause of eclipses ★★	★ Explained that eclipses are caused by SHADOWS — the Moon entering Earth's shadow (lunar) and the Moon's shadow falling on Earth (solar). He explicitly rejected the Rahu-Ketu demon explanation
Moon shines by reflected light ★	Correctly stated
Value of π ★	Gave $\pi \approx 3.1416$, and remarkably described it as <i>asanna</i> — "approaching" / approximate. ★ He appears to have understood that π is irrational
Circumference of the Earth ★	Calculated as ~39,968 km — the modern value is ~40,075 km. Accurate to within 0.3%
Length of the year	~365.358 days
Trigonometry	Sine tables (<i>jya</i>) — the word <i>jya</i> → Arabic <i>jiba</i> → Latin <i>sinus</i> → " sine " ★
Algebra	Solutions to indeterminate equations (<i>kuttaka</i> method)

★ **The *asanna* detail is the one to remember.** Aryabhata didn't just compute π to four places — he flagged the value as an **approximation**, indicating an understanding that no exact fraction exists. **That is mathematical sophistication of a very high order, and it's a memorable, specific fact for an answer.**

🌐 India's **first satellite (1975) was named Aryabhata.**

★★★ ZERO and the Decimal Place-Value System

India's greatest single gift to the world.

Element	Detail
The concept	Zero (<i>shunya</i>) as both a placeholder and a number in its own right
Place value	Digits' values depend on position — 2 in "20" ≠ 2 in "200"
Earliest written zero ★	The BAKSHALI MANUSCRIPT (a dot used as zero; carbon-dated portions as early as the 3rd–4th century CE) and the GWALIOR INSCRIPTION (876 CE) , which has the earliest confirmed inscriptional zero shaped like our "0"
BRAHMAGUPTA (598–668 CE) ★★	In the <i>Brahmasphutasiddhanta</i> (628 CE), he gave the first systematic RULES FOR ARITHMETIC WITH ZERO and with NEGATIVE NUMBERS ★ — $a+0=a$, $a \times 0=0$, and rules for debts (negatives) and fortunes (positives). Also the Brahmagupta formula for cyclic quadrilaterals
Transmission ★	India → Arab world (via Sindh contact from 712 CE and Al-Khwarizmi's works) → Europe (via Fibonacci, 1202). Hence "Arabic numerals" in Europe, but <i>al-arqam al-hindiyyah</i> ("Indian numerals") in Arabic



Take a second with this one, because it is genuinely yours.

Without positional notation and zero:

- There is no efficient arithmetic (try long division in Roman numerals)
- There is no algebra as we practise it
- There is **no binary representation** — binary is positional notation in base 2, and it needs a symbol for "nothing in this position"
- There is **no computing**

Every `int`, every memory address, every bitmask, every `0` you have ever typed rests on a conceptual leap made in ancient India: **treating "nothing" as a thing you can write down and calculate with.**



This is superb Essay material on India's scientific heritage — and unlike most such claims, it is uncontested and precisely documented.

Other mathematicians and astronomers

Figure	Contribution
VARAHAMIHIRA (6th c.) ★	Brihat Samhita (an encyclopaedia — astronomy, architecture, gemology, agriculture), Panchasiddhantika (summarising five astronomical systems, including Greek). Traditionally a "navaratna"
BRAHMAGUPTA (7th c.) ★★	Brahmasphutasiddhanta — zero and negative numbers
BHASKARA I (7th c.)	First to write numbers in the Hindu decimal system with a circle for zero; commentary on Aryabhata
BHASKARA II / Bhaskaracharya (12th c.) ★	Siddhanta Shiromani , including Lilavati (arithmetic) and Bijaganita (algebra); early ideas approaching calculus
BAUDHAYANA ★	Sulbasutras — geometry for altar construction; ★ contains a statement of what is now called the Pythagorean theorem, predating Pythagoras
Aryabhata II, Madhava of Sangamagrama	Madhava (14th c., Kerala School) developed infinite series for π and trigonometric functions — anticipating aspects of calculus



Medicine ★★

Figure	Work	Known as
CHARAKA ★★	CHARAKA SAMHITA	★ MEDICINE . The foundational text of Ayurveda ; diagnosis, physiology, ethics of the physician, pharmacology. Associated with Kanishka's court (Ch.8)
SUSHRUTA ★★	SUSHRUTA SAMHITA	★ SURGERY . Called the ★ "Father of Surgery." Describes ~ 300 surgical procedures and 120 instruments , including ★ RHINOPLASTY (nose reconstruction) and cataract surgery
Vagbhata	<i>Ashtanga Hridaya</i>	The third of the "great triad" of Ayurveda
Nagarjuna	—	Alchemy and metallurgy (distinct from the Mahayana philosopher of the same name)

★ **Sushruta's rhinoplasty is a genuinely remarkable, specific fact:** he describes reconstructing a nose using a **flap of skin from the cheek or forehead** — a technique that reached Europe in the 18th century (via a report from India published in 1794) and is **still called the "Indian method"** in reconstructive surgery. **An outstanding example for a Science & Tech or Essay answer.**

⚙️ Metallurgy — the Iron Pillar ★★ ★

The MEHRAULI IRON PILLAR, Qutub Minar complex, Delhi

Fact	Detail
When	★ ~4th–5th century CE (Gupta period)
Who	Inscription credits a king ★ "CHANDRA" — generally identified with CHANDRAGUPTA II
Height	~7.2 m (about 24 feet)
Weight	~6 tonnes (often cited as ~3 tonnes for the shaft; ~6 tonnes total)
★★ THE MARVEL	It has not rusted through in over 1,600 years , standing exposed in the Delhi open air
Why? ★	Modern analysis (notably by R. Balasubramaniam, IIT Kanpur) shows a high phosphorus content (~1%) and the absence of sulphur/magnesium, which — with the alternating wet/dry Delhi climate — formed a thin protective passive layer of "misawite" (iron hydrogen phosphate hydrate) over the surface, sealing it against further corrosion
Also demonstrates	The ability to forge-weld a 6-tonne iron mass — an extraordinary feat for the period

🎯 **How to phrase this well:** *"The Mehrauli pillar demonstrates not accidental durability but sophisticated control of iron chemistry — the high phosphorus content that produced its protective passive layer reflects deliberate metallurgical practice, and it stands as evidence of the technical capability of Gupta-era ironworking."*

⚠️ **Avoid the mystical framing** ("mysteriously never rusts"). The mechanism is understood. **Explaining it correctly is more impressive than treating it as a mystery.**

Also in metallurgy: the **Sultanganj Buddha** (Bihar, Gupta period) — a **2.3 m, ~500 kg copper Buddha**, the largest surviving ancient Indian metal image; and India's ancient reputation for **wootz steel**, exported west and famed as the basis of "Damascus steel."

🏆 UNESCO World Heritage Sites from This Book ★★

A high-frequency Prelims and interview area. Know these.

Site	Inscribed	Chapter
Ajanta Caves	1983	Ch.10
Ellora Caves	1983	Ch.10
Sanchi Buddhist Monuments	1989	Ch.7, 10
Mahabodhi Temple, Bodh Gaya	2002	Ch.6, 10
Bhimbetka Rock Shelters	2003	Ch.2
Group of Monuments at Mahabalipuram	1984	Ch.9, 10
Group of Monuments at Pattadakal	1987	Ch.9, 10
Great Living Chola Temples	1987/2004	Ch.9, 10
Elephanta Caves	1987	Ch.10
Dholavira: A Harappan City	2021	Ch.3
Nalanda Mahavihara	2016	Ch.9
Lumbini (in Nepal)	1997	Ch.6

Prelims

Topic	The exact thing to know
Stupa parts ★★ ★	Anda, harmika, yashti, chhatri, medhi, pradakshina patha, vedika, torana
Sanchi / Bharhut / Amaravati ★★ ★	Ashoka→Shunga→Satavahana; earliest narrative relief; white marble + Satavahana
Aniconic symbols ★★ ★	Empty throne, footprints, bodhi tree, wheel, stupa, riderless horse
Chaitya vs Vihara ★★ ★	Prayer hall (apsidal, stupa, barrel vault) vs monastery (cells)
Karle	Largest and finest chaitya
Ajanta ★★ ★	30 caves; paintings; TEMPERA, not fresco; Padmapani & Vajrapani (Cave 1); Satavahana + Vakataka ; UNESCO
Ellora ★★ ★	34 caves; Buddhist + Hindu + Jain; Cave 16 Kailashnatha; Rashtrakuta Krishna I
Kailashnatha ★★ ★	Monolithic, carved top-down , ~200,000 tonnes removed
Temple parts ★★ ★	Garbhagriha, shikhara, vimana, mandapa, gopuram, amalaka
Nagara vs Dravida ★★ ★	Curvilinear shikhara + amalaka, no boundary wall vs stepped vimana + gopuram + boundary wall + tank
Mahabalipuram ★★ ★	Pancha Rathas (monolithic, five different styles), Shore Temple (structural), Descent of the Ganga
Pallava sequence ★	Cave → monolith (ratha) → structural
Aihole / Pattadakal ★★ ★	Cradle of temple architecture / UNESCO, both styles side by side
Gupta temples ★	Sanchi 17, Tigawa, Deogarh, Bhitargaon (brick) — earliest structural temples
Epics ★	<i>Mahabharata</i> — Vyasa , ~100,000 verses, contains the Gita ; <i>Ramayana</i> — Valmiki
Panchatantra ★★ ★	Vishnu Sharma ; most-translated
Kalidasa ★★ ★	<i>Abhijnana Shakuntalam</i> , <i>Meghaduta</i> , <i>Raghuvamsha</i> , <i>Kumarasambhava</i> , <i>Ritusamhara</i>
Natyashastra ★★ ★	Bharata ; drama, dance, music; rasa theory
Mrichchhakatika	Shudraka
Aryabhata ★★ ★	<i>Aryabhatiyam</i> ; Earth rotates; eclipses by shadow; $\pi \approx 3.1416$ (asanna) ; Earth's circumference
Brahmagupta ★★ ★	<i>Brahmasphutasiddhanta</i> (628 CE); rules for zero and negative numbers
Varahamihira ★	<i>Brihat Samhita</i> , <i>Panchasiddhantika</i>
Baudhayana ★	<i>Sulbasutras</i> ; Pythagorean theorem statement
Charaka / Sushruta ★★ ★	Medicine / Surgery ("Father of Surgery," rhinoplasty)
Mehrauli Iron Pillar ★★ ★	~7.2 m, king " Chandra " (Chandragupta II), rust-resistant due to high phosphorus
Zero ★★ ★	Bakhshali manuscript, Gwalior inscription (876 CE)
UNESCO sites ★★ ★	The table above

Traps:

- "Ajanta paintings are frescoes" → **FALSE**. **Tempera** on dry plaster
- "Ellora is only Buddhist" → **FALSE**. **Buddhist, Hindu AND Jain**
- "Kailashnatha was built stone by stone" → **FALSE**. **Carved from a single rock, top-down**
- "Gopurams are found in Nagara temples" → **FALSE**. **Dravida**

- "In Dravida temples, the tower over the sanctum is called the shikhara" → **FALSE**. It is the **vimana**; "shikhara" is only its crowning element
- "Sushruta wrote on medicine and Charaka on surgery" → **FALSE**. The reverse
- "Ajanta has both Hindu and Buddhist caves" → **FALSE**. Ajanta is **entirely Buddhist**; Ellora is the multi-religious one
- "Aryabhata said the Sun revolves around the Earth" → He stated **the Earth rotates on its axis**; be careful with heliocentrism claims

Mains — this is the chapter that answers the syllabus line

Very high-probability question types:

1. "Discuss the salient features of Indian rock-cut architecture." → Barabar → Bhaja/Karle chaityas → Ajanta → Ellora → Mahabalipuram rathas
2. "Trace the evolution of temple architecture in India." → Gupta flat-roofed shrines → shikhara emerges → **Nagara/Dravida/Vesara** → the Pallava **cave**→**monolith**→**structural** sequence → Chola
3. "Early Buddhist stupa-art, while depicting folk motifs and narratives, successfully expounds Buddhist ideals." (2016) → Sanchi/Bharhut toranas, Jatakas in stone, **aniconic symbolism**, folk yakshi figures absorbed into Buddhist settings
4. "Discuss the main contributions of the Gupta period and the Chola period to Indian heritage and culture." (2022) → Gupta: Kalidasa, Aryabhata, Ajanta, temple beginnings, iron pillar, gold coins. Chola: Brihadeshwara, bronze **Nataraja**, local self-government, naval reach
5. "Ancient India made significant contributions to science and mathematics." → zero, decimal place value, Aryabhata, Brahmagupta, Sushruta, metallurgy
6. **Ellora as evidence of religious coexistence** — deployable in any answer on Indian pluralism

💡 **A technique that works for Art & Culture answers: draw a small labelled diagram.** A quick sketch of a stupa or a temple with parts labelled takes 90 seconds, demonstrates command of terminology, and breaks up the text. Examiners consistently reward it.

Glossary

Term	Meaning
Stupa	Mound enshrining a relic
Anda	The hemispherical dome ("egg")
Harmika	Square railing atop the dome
Yashti	Central mast
Chhatra	Umbrella tiers of honour
Medhi	Drum/base of the stupa
Pradakshina patha	Circumambulatory path
Vedika	Stone railing
Torana	Carved gateway
Chaitya	Buddhist prayer hall (apsidal, with a stupa)
Vihara	Monastery with monks' cells
Gavaksha	Horseshoe-shaped chaitya window
Garbhagriha	"Womb-house" — temple sanctum
Shikhara	Tower over the sanctum (Nagara) / crowning element (Dravida)
Vimana	Pyramidal tower over the sanctum (Dravida)
Mandapa	Pillared hall
Antarala	Vestibule
Gopuram	Gateway tower (Dravida)
Amalaka	Ribbed disc crowning a Nagara shikhara
Kalasha	Pot-shaped finial
Nagara / Dravida / Vesara	Northern / Southern / Deccan hybrid temple styles
Monolithic	Carved from a single rock
Tempera	Pigment with a binder on dry plaster (Ajanta)
Fresco	Pigment on wet plaster (not Ajanta)
Ratha	Monolithic temple at Mahabalipuram ("chariot")
Rasa	Aesthetic emotion/flavour (<i>Natyashastra</i>)
Shunya	Zero
Asanna	"Approaching" — Aryabhata's term for his approximate π
Sulbasutras	Texts of altar geometry (Baudhayana)

2-MINUTE RECAP

- Stupa parts:** **anda** (dome with relic), **harmika**, **yashti**, **chhatra**, **medhi**, **pradakshina patha**, **vedika**, **torana**. Worship is by **walking around**, so the carvings unfold as narrative.
- Sanchi** (Ashoka → Shunga → Satavahana toranas; UNESCO), **Bharhut** (earliest narrative relief), **Amaravati** (white marble, Satavahana, influenced SE Asia).

3. ★ **Aniconic phase:** the Buddha shown by **empty throne, footprints, bodhi tree, wheel, stupa, riderless horse** — human images only from the 1st century CE.
4. **Chaitya** = apsidal prayer hall with a stupa and barrel vault (**Karle** the finest). **Vihara** = monastery with cells. Stone ribs imitate **lost wooden architecture**.
5. **Ajanta:** 30 caves, Buddhist only, **paintings in TEMPERA (not fresco)**, Padmapani & Vajrapani, Satavahana + **Vakataka**, UNESCO.
6. **Ellora:** 34 caves, ★ **Buddhist + Hindu + Jain together**, **Cave 16 = Kailashnatha**, Rashtrakuta Krishna I, ~200,000 tonnes removed, **carved top-down from one rock**.
7. **Temple parts:** **garbhagriha, shikhara/vimana, mandapa, gopuram, amalaka**.
8. **Nagara** = curvilinear shikhara + amalaka, no boundary wall. **Dravida** = stepped vimana + **gopuram** + boundary wall + tank. **Vesara** = Deccan hybrid.
9. **Mahabalipuram: Pancha Rathas** (monolithic, five styles), **Shore Temple** (structural), **Descent of the Ganga**. **Pallava sequence: cave → monolith → structural**.
10. **Aihole** = cradle of temple architecture; **Pattadakal** = UNESCO, both styles side by side. **Gupta temples** (Sanchi 17, Tigawa, Deogarh, Bhitargaon) are the earliest free-standing ones.
11. **Literature:** *Mahabharata* (**Vyasa**, longest poem, contains the **Gita**), *Ramayana* (**Valmiki**), **Puranas** (open to all, unlike the Vedas), **Jatakas**, **Panchatantra** (**Vishnu Sharma**, most-translated).
12. **Kalidasa:** *Abhijnana Shakuntalam*, *Meghaduta*, *Raghuvamsha*, *Kumarasambhava*, *Ritusamhara*. **Bharata's Natyashastra** — the basis of Indian classical dance and the **rasa** theory.
13. **Aryabhata:** **Earth rotates; eclipses by shadow, not Rahu; $\pi \approx 3.1416$** flagged as **asanna** (approximate); Earth's circumference accurate to 0.3%.
14. **Brahmagupta (628 CE):** first systematic **rules for zero and negative numbers**.
15. ★ **Zero + decimal place value** travelled India → Arabs → Europe. **Every computation you have ever run depends on it.**
16. **Charaka = medicine; Sushruta = surgery** ("Father of Surgery," **rhinoplasty**, ~300 procedures).
17. **Mehrauli Iron Pillar:** ~7.2 m, king "**Chandra**" (Chandragupta II), **rust-resistant due to high phosphorus** forming a passive layer — understood chemistry, not mystery.
18. **Know the UNESCO list** — Ajanta, Ellora, Sanchi, Mahabodhi, Bhimbetka, Mahabalipuram, Pattadakal, Chola temples, Elephanta, **Dholavira (2021), Nalanda (2016)**.

Test Yourself

Prelims style

1. Consider the following about the Ajanta caves:
 1. They contain Buddhist, Hindu and Jain caves.
 2. The paintings were executed in the tempera technique.
 3. They were patronised in part by the Vakatakas. → **2 and 3 only. Ellora is the multi-religious site; Ajanta is entirely Buddhist.**
2. The Kailashnatha temple at Ellora —
 1. was built by the Rashtrakuta ruler Krishna I.
 2. was carved from a single rock, working downward from the top.
 3. is Dravidian in architectural idiom. → **All three.**
3. Which of the following is/are characteristic of the Dravida style?
 1. A boundary wall enclosing the temple complex

2. A gopuram over the entrance
3. An amalaka crowning the tower → 1 and 2 only. The **amalaka** is a **Nagara** feature.
4. In early Buddhist art, the Buddha's first sermon was represented by — → *The wheel (dharmachakra).*
5. Consider the following pairs:
 1. Charaka Samhita — surgery
 2. Sushruta Samhita — medicine
 3. Aryabhatiyam — astronomy and mathematics → Only 3. Charaka = **medicine**, Sushruta = **surgery** — the pairs are reversed.
6. The Mehrauli Iron Pillar's resistance to corrosion is principally attributed to — → *Its high phosphorus content, which formed a protective passive layer.*
7. The Pancha Rathas at Mahabalipuram are notable because — → *Each is monolithic and each is in a different architectural style.*

Mains style

8. Discuss the salient features of Indian rock-cut architecture from the Mauryan to the Rashtrakuta period. **(250 words)**
9. Trace the evolution of temple architecture in India, distinguishing the Nagara and Dravida styles. **(250 words)**
10. "Early Buddhist stupa-art, while depicting folk motifs and narratives, successfully expounds Buddhist ideals." Elucidate. **(150 words) — UPSC 2016**
11. Discuss the main contributions of the Gupta period and the Chola period to Indian heritage and culture. **(250 words) — UPSC 2022**
12. Assess the contribution of ancient India to mathematics and astronomy. **(250 words)**

Self-check (out loud, no notes):

- Draw and label a stupa. Then draw and label a temple.
 - Name six aniconic symbols and what each represents.
 - Give four differences between a chaitya and a vihara.
 - Why is Ellora more remarkable than Ajanta in one specific respect — and vice versa?
 - Give five differences between Nagara and Dravida.
 - Name Kalidasa's five major works.
 - State four of Aryabhata's contributions.
 - Why doesn't the Mehrauli pillar rust?
-

10 chapters · 2 million years · from stone tools to zero

WHAT NEXT:

1. Revise using the six THREADS in the Index – narrate each one aloud without notes. If you can, you've understood it.
2. Take the Class 6 History quiz (05_Quiz_MCQ). Target 85%+.
3. Move to Class 7 (Our Pasts II – Medieval India).

Remember the Index's warning: Ancient India is high-effort, moderate-return. You now have the foundation. Don't over-invest here – Modern India is worth 3-4× as much.